Source-Driven Global–Local Prototype Contrastive Learning for Hyperspectral Domain Adaptation

Yuhao Xie, Ronghua Shang, *Senior Member, IEEE*, Jinhong Ren, Jingyu Zhong, Jie Feng, *Senior Member, IEEE*, and Songhua Xu

***Abstract*—Hyperspectral domain adaptation (HSDA) recontextualizes unsupervised domain adaptation (UDA) for the spectrally rich and structurally complex task of hyperspectral image classification (HSIC). However, most existing HSDA methods confine source supervision to an auxiliary role for stabilizing training, overlooking its potential to act as a central driving force in the adaptation process. Moreover, global alignment at the domain level has not been effectively emphasized in existing methods. To address these limitations, we propose a source-driven global–local prototype contrastive learning (SGLPCL) framework that fully leverages source supervision as a central driver of adaptation through hierarchical prototype alignment, fostering both domain-invariant and semantically coherent representations. To this end, SGLPCL constructs robust node representations from spectral–spatial–semantic embeddings and learns global representations through a multihead self-attention graph network. It then refines semantic pseudolabels at both global and local levels using multiscale distance metrics, leveraging source supervision for reliable labels. Finally, a global–local prototype contrastive learning framework is established, with an adaptive temperature scaling (ATS) matrix, utilizing source supervision to calibrate prototype similarities and enhance contrastive learning robustness and discriminability. Comprehensive evaluations across three representative hyperspectral benchmarks substantiate the superiority of SGLPCL, which attains remarkable improvements in accuracy and Kappa coefficient over ten leading methods.**

***Index Terms*—Global alignment, hyperspectral image classification (HSIC), prototype contrastive learning, source supervision, unsupervised domain adaptation (UDA).**

Received 6 February 2026; accepted 10 March 2026. Date of publication 17 March 2026; date of current version 3 April 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62576260, Grant 62176200, Grant 12326617, and Grant 62271374. *(Corresponding author: Ronghua Shang.)*

Yuhao Xie is with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Guangzhou Institute of Technology, Xidian University, Guangzhou, Guangdong 510000, China (e-mail: yuhaoxie@stu.xidian.edu.cn).

Ronghua Shang, Jinhong Ren, Jingyu Zhong, and Jie Feng are with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, and the School of Artificial Intelligence, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: rhshang@mail.xidian.edu.cn; joneren@stu.xidian.edu.cn; zhongjingyu2001@163.com; jiefeng@xidian.edu.cn).

Songhua Xu is with the Department of Health Management and the Institute of Medical Artificial Intelligence, The Second Affiliated Hospital of Xi'an Jiaotong University, Xi'an 710049, China (e-mail: songhuaxu@mail.xjtu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3674804

NOMENCLATURE

Z_s, Z_t	Local feature representations for source/target domains.
G_s, G_t	Global feature representations for source/target domains.
$\hat{Y}, \check{Y}$	Local target pseudolabels, and global target pseudolabels.
M_{sl}, M_{tl}	Memory pools for source-/target-domain local features.
M_{sg}, M_{tg}	Memory pools for source-/target-domain global features.

I. INTRODUCTION

HYPERSPECTRAL image classification (HSIC) seeks to assign a land cover label to each pixel. It plays a critical role in remote sensing, underpinning informed decision-making in applications such as environmental monitoring, natural resource management, and urban planning [1]. Recently, deep learning techniques, including convolutional neural networks [2], deep encoding [3], and attention-based architectures [4], have demonstrated impressive performance in HSIC by effectively modeling spectral–spatial dependencies. Nonetheless, these approaches require extensive annotated data, which is expensive and labor-intensive to acquire [5]. Concurrently, hyperspectral datasets often encompass abundant unlabeled samples that remain largely underexploited [6].

Unsupervised domain adaptation (UDA) has emerged as a promising paradigm to reduce reliance on labeled data by transferring discriminative knowledge from a labeled source domain to an unlabeled target domain [7], [8]. This learning paradigm is particularly relevant to HSIC, where unlabeled samples are abundant but annotated data are extremely limited. However, most UDA algorithms are designed for natural RGB images [9], [10], [11], and their assumptions about data distribution, feature structure, and modality characteristics do not hold for hyperspectral data. Specifically, hyperspectral images (HSIs) exhibit high spectral dimensionality, strong spatial correlations, and significant spectral redundancy, which together complicate feature alignment and amplify interdomain discrepancies [12]. Consequently, these methods struggle to capture the intricate spectral–spatial dependencies essential for hyperspectral understanding, leading to unstable adaptation and limited transferability. These challenges highlight the necessity of developing dedicated hyperspectral domain

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

adaptation (HSDA) frameworks that explicitly consider hyperspectral characteristics and effectively reduce domain shifts under label-scarce conditions [13], [14].

To this end, recent HSDA research has explored various strategies to alleviate cross-domain discrepancies and enhance the generalization of hyperspectral classifiers. Early studies have predominantly focused on distribution alignment to mitigate domain shifts. Distance-based methods [14] explicitly minimize statistical divergence between domains using measures such as maximum mean discrepancy (MMD) [15]. Building upon this idea, Zhang et al. [16] propose a topological structure and semantic transfer network, which performs explicit distribution alignment via MMD and further aligns topological structures through graph-based optimal transport, effectively alleviating interdomain spectral shifts. In parallel, adversarial learning frameworks [17], [18] introduce implicit alignment by training a feature extractor and a domain discriminator through minimax optimization [19], to learn domain-invariant feature representations that generalize across domains. Building upon this paradigm, Tang et al. [20] proposed a feature extractor–domain discriminator–dual classifier adversarial framework, in which the extractor and discriminator align cross-domain spectral statistics, while dual classifiers maximize output discrepancy on unlabeled target data to produce embeddings that respect source decision boundaries and interclass separability, jointly addressing spectral shift and discriminative structure preservation. Though these approaches reduce interdomain bias, they often compromise class discriminability due to excessive feature alignment and the absence of semantic constraints.

Enhancing the discriminability of domain-invariant features has motivated recent HSDA studies to increasingly employ self-training [21], [22] as an effective adaptation strategy. By leveraging pseudolabeled target samples, self-training augments supervision and progressively refines the decision boundary between source and target domains [23], thereby promoting intraclass compactness and interclass separability for more discriminative representations [24]. Wang et al. [25] further employed a dual-classifier collaboration strategy to filter high-confidence pseudo labels and iteratively expand reliable target samples, while incorporating spectral band perturbation to enhance cross-domain hyperspectral feature alignment. However, the effectiveness of self-training is often hindered by noisy or inaccurate pseudo labels [26], which may propagate erroneous supervision and impose rigid negative constraints on target features, ultimately limiting model robustness. To mitigate these issues, contrastive learning frameworks [27], [28] have emerged as a promising alternative. By optimizing the relative distances between positive and negative pairs, contrastive learning alleviates the impact of incorrect pseudo labels while enhancing feature separability and structural consistency [29]. This contrastive learning paradigm facilitates more reliable and discriminative domain-invariant representations, providing a robust pathway toward effective cross-domain hyperspectral adaptation. Within this broader framework, Cai et al. [30] demonstrated how supervised contrastive learning guided by target-domain pseudo labels can bring together spectral–spatial features of the same class across source and target domains while pushing apart those of different classes, thereby improving both the discriminability and domain invariance of cross-scene hyperspectral features.

Although considerable progress has been made in HSDA, most studies focus on improving target-domain pseudo labels in unsupervised settings, while the fully reliable supervision from the source domain remains largely underexplored. Due to the domain gap between source and target distributions, source labels are often relegated to a passive and auxiliary role, typically serving only to stabilize the adaptation process through conventional cross-entropy (CE) losses on source data. However, source annotations inherently contain rich and trustworthy semantic information that could be leveraged as an active driving force for domain adaptation, rather than being confined to a mere stabilizing constraint. Furthermore, current HSDA approaches largely emphasize local feature alignment at the level of individual pixel (or superpixel) patches or class-wise distribution matching [31], [32], while paying limited attention to global contextual consistency across domains. Considering the complex spectral–spatial correlations and hierarchical dependencies inherent in HSIs, inadequate holistic alignment weakens the preservation of spectral structures and compromises global semantic coherence.

To address the aforementioned limitations, we propose an source-driven global–local prototype contrastive learning (SGLPCL) framework that fundamentally redefines the role of source supervision in the HSDA task. Unlike previous HSDA methods that treat source labels as passive auxiliary constraints for stabilizing training, SGLPCL reimagines source labels as an active driving force throughout the adaptation process, manifested through three pivotal innovations that drive global representation learning, enable pseudolabel refinement, and facilitate prototype contrastive learning. Specifically, SGLPCL encodes spectral–spatial–semantic cues into node representations and utilizes a multihead self-attention graph network to capture global contextual representations of HSIs, ensuring a holistic and seamless cross-domain alignment. Differing from prior methods that coarsely estimate global features at the pixel or superpixel level, SGLPCL integrates holistic domain-level information into global representations, enabling more comprehensive and semantically coherent feature encoding across domains. To further leverage source supervision, SGLPCL identifies semantically correlated source neighbors at global and local scales through a multidistance metric, adaptively incorporating accurate source labels to mitigate noise in pseudolabels compared with previous unsupervised refinement methods. Finally, SGLPCL constructs source-guided class prototypes by fusing pseudolabeled target features with source representations, which are refined dynamically via an adaptive temperature matrix computed from source labels. This adaptive mechanism ensures stable contrastive optimization and stronger interclass separability across domains, addressing the issues of prototype drift and intraclass dispersion observed in existing contrastive learning methods. The main contributions are summarized as follows.

1) A spectral–spatial–semantic node construction mechanism is developed to enable global representation learning. By embedding 3-D spectral, spatial, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

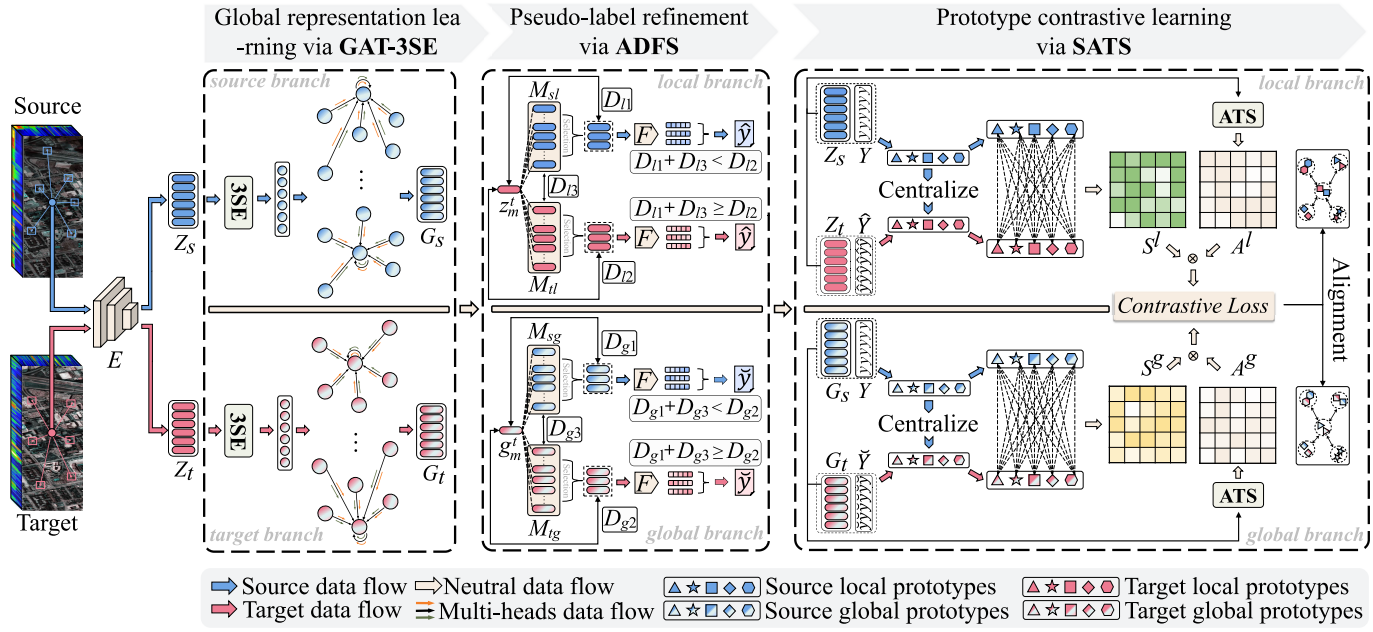


Fig. 1. Overview of the proposed SGLPCL framework. It mainly consists of three stages. First, GAT-3SE learns global representations (G_s, G_t) within the source (upper part) and target domain (lower part). On this basis, ADFS adaptively refines pseudolabels ($\check{y}$, $\hat{y}$) for each unlabeled global (lower part) and local (upper part) target-domain feature by leveraging similar candidates from the global (M_{sg}, M_{tg}) and local (M_{sl}, M_{tl}) memory pool and their multiscale distances (D_{g1-3}, D_{l1-3}) to assess matching reliability and determine the final pseudolabel generation strategy. Finally, leveraging the learned representations (e.g., G_s, G_t) and refined pseudolabels ($\check{Y}$, $\hat{Y}$) from previous stages, SATS adaptively updates class prototypes and performs temperature-calibrated prototype contrastive learning at both global (lower part) and local (upper part) levels to achieve robust cross-domain alignment.

semantic cues into node representations, domain-level contextual features are captured by the multihead self-attention graph network, facilitating coherent and holistic alignment across source and target domains.

2) A source-guided adaptive domain feature selection (ADFS) strategy is proposed for pseudolabel refinement. By leveraging source supervision, semantically correlated neighbors across global and local scales are identified using a multidistance metric, effectively refining and stabilizing target pseudolabels for reliable adaptation.
3) A source-centroid and temperature-calibrated mechanism is introduced to facilitate prototype contrastive learning. Source supervision drives the construction of global–local hierarchical class prototypes by integrating pseudolabeled target features with source representations. These prototypes are dynamically refined via adaptive temperature scaling (ATS), enhancing cross-domain discriminability and ensuring invariance.

This article is structured as follows. Section II provides a detailed description of the proposed algorithm. Section III presents experimental results on three benchmark datasets. Finally, Section IV summarizes the conclusions of this article.

II. Proposed Method

To fully harness the reliable supervision from the source domain and ensure coherent global–local alignment in HSDA, we propose the SGLPCL framework, explicitly leveraging the rich semantic information in source annotations and addressing the limitations of existing methods in capturing global contextual consistency. SGLPCL follows the standard HSDA setting, with access to source samples $X_s = \{x_m^s\}_{m=1}^{n_s}$ with labels $Y = \{y_m\}_{m=1}^{n_s}$ and unlabeled target samples $X_t = \{x_m^t\}_{m=1}^{n_t}$, where n_s and n_t denote the number of source and target samples. As illustrated in Fig. 1, the feature extractor E of SGLPCL initially maps X_s and X_t to local feature representations (i.e., representation vectors) $Z_s = \{z_m^s\}_{m=1}^{n_s}$ and $Z_t = \{z_m^t\}_{m=1}^{n_t}$, which are feature embeddings rather than classifier logits. SGLPCL then consistently refines domain adaptation through three interdependent stages. To enable global–local prototype contrastive learning in the absence of global representations and target-domain labels, we first employ the graph attention network with spatial-spectral–semantic embeddings (GAT-3SE). This facilitates the derivation of global representations that incorporate essential domain-level knowledge. With this foundation, ADFS injects source supervision to distill high-confidence pseudolabels from the memory pool, thereby effectively guiding unlabeled global and local target features toward more robust prototype aggregation. Finally, leveraging the learned global–local representations and refined pseudolabels, SGLPCL performs global–local prototype contrastive learning via source-centroid and ATS (SATS). This stage constructs source-guided prototypes by fusing labeled source and pseudolabeled target features, and dynamically adjusts the temperature scaling matrix to balance interclass separability and intraclass compactness under cross-domain contrastive loss. To facilitate reference, key symbols of SGLPCL are defined in Nomenclature.

A. Global Representation Learning via GAT-3SE

Unlike prior methods [31], [32] that coarsely estimate global features at the pixel or superpixel level, we propose GAT-3SE to fuse holistic domain-level information into global

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

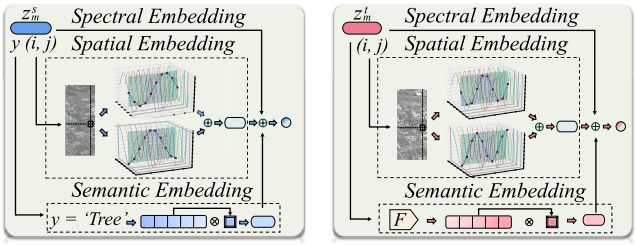


Fig. 2. 3SE node construction diagram. ⊕ denotes channel-wise feature concatenation. ⊗ denotes multiplicative feature interaction. (a) Source node. (b) Target node.

representations. In essence, GAT-3SE converts local features from both domains into expressive graph nodes through the 3SE embedding mechanism, which structurally couples spectral–spatial cues with semantic embeddings, yielding representations that are both locally descriptive and semantically coherent. Built upon these semantically grounded nodes, a graph attention network is employed to perform domain-wide relational modeling, allowing informative nodes to be adaptively attended to and aggregated across the entire domain. Through this global message passing and attention-driven aggregation process, GAT-3SE derives source- and target-domain global feature representations that encode holistic, domain-level characteristics beyond local neighborhoods. The workflow of the 3SE is illustrated in Fig. 2.

From Fig. 2, each local feature z_m (z_m^s or z_m^t), rich in spectral information, acts as the spectral embedding e_{spe} for the local nodes: $e_{spe} = z_m$. Moreover, spatial embedding e_{spa} is incorporated by encoding the 2-D pixel coordinate (i, j), enhancing spatial relationships among nodes utilizing multifrequency sine and cosine functions [33]

$$\begin{aligned} e_{spa}[:, 2k] &= \sin\left(i \times 1/10000^{2k/d_{model}}\right) \\ e_{spa}[:, 2k+1] &= \cos\left(i \times 1/10000^{2k/d_{model}}\right) \end{aligned} \tag{1}$$

where d_{model} is the feature length of e_{spa}, and k is the dimension index, ranging from 0 to $d_{model}/4$. The encoding for j is similar: simply replace i with j in (1) and adjust the starting index to $d_{model}/2$. To enhance semantic relationship discernment, individual nodes embed rich and discriminative class semantic representations internally. For the target node, the local feature z_m^t is predicted by classifier F and SoftMax-normalized (sm) into a classifier probability $p = \mathrm{sm}(F(z_m^t))$, with confidence $\mathcal{C}(p)$ obtained by linearly transforming its entropy $\mathcal{H}(p)$: $\mathcal{C}(p) = (\mathcal{H}_{max} - \mathcal{H}(p))/(\mathcal{H}_{max} - \mathcal{H}_{min})$. Here, $\mathcal{H}(p)$ can be calculated as in [34], with $\mathcal{H}_{max} = \log c$ and $\mathcal{H}_{min} = 0$. c is the total number of classes. Finally, the semantic embedding for the target node is $e_{sem} = p \times \mathcal{C}(p)$. The semantic embedding of the source node follows the same process as the target node, except that the probability p is directly obtained by converting the label y_m into a one-hot vector instead of relying on the classifier output. By concatenating e_{spe}, e_{spa}, and e_{sem}, the strongly representative node h_m for the local feature z_m is obtained

$$h_m = e_{spe} \parallel e_{spa} \parallel e_{sem} \tag{2}$$

where $\parallel$ represents feature concatenation. By applying the 3SE mechanism to construct nodes for each local feature in Z_s and Z_t, the source and target node sets, $H_s = \{h_m^s\}_{m=1}^{n_s}$ and $H_t = \{h_m^t\}_{m=1}^{n_t}$, are derived.

Next, a multihead graph attention network facilitates node interaction within H_s and H_t, promoting the learning of global representations. Specifically, the global representation g_m^s is derived through information aggregation centered on the source node h_m^s as follows:

$$g_m^s = \Big\Vert_{k=1}^{K} \sigma\left(\sum_{v \in \mathcal{N}_m} \alpha_{mv}^k W h_v^s\right) \tag{3}$$

where $\Big\Vert$ denotes concatenation, K is the number of attention heads, $\sigma(\cdot)$ is the nonlinear activation, and W is the shared linear transformation weight. $\mathcal{N}_m$ is the set of neighbors for node h_m^s, and h_v^s is one such neighbor. Neighbor selection follows [16], using multiround random subgraph construction to balance computational cost and full node interaction. α_{mv}^k denotes the correlation of h_m^s and h_v^s under the kth attention head. As in [35], node correlations are computed using a single-layer feedforward network with LeakyReLU activation and normalized to obtain a_{mv}^k

$$\alpha_{mv}^k = \frac{\exp\left(\mathrm{LeakyReLU}\left(w_k^{\mathrm{T}}\left(W h_m^s \| W h_v^s\right)\right)\right)}{\sum_{u \in \mathcal{N}_m} \exp\left(\mathrm{LeakyReLU}\left(w_k^{\mathrm{T}}\left(W h_m^s \| W h_u^s\right)\right)\right)} \tag{4}$$

where w_k^{T} denotes the transpose of the feedforward network weight. Each node in H_s and H_t undergoes multihead attention, yielding the global representations $G_s = \{g_m^s\}_{m=1}^{n_s}$ and $G_t = \{g_m^t\}_{m=1}^{n_t}$ for the source and target domains.

B. Pseudolabel Refinement via ADFS

Previous pseudolabel-based methods [21], [23] typically refine pseudolabels within a fully unsupervised framework, often introducing significant noise that undermines label reliability. In contrast, ADFS explicitly integrates accurate source-domain supervision during the refinement process, thereby significantly enhancing the reliability of the refined pseudolabels. As it operates similarly on both global and local features, we describe only the global refinement for clarity.

Two global memory pools, $M_{sg} = \{\tilde{g}_u^s\}_{u=1}^{n}$ and $M_{tg} = \{\tilde{g}_u^t\}_{u=1}^{n}$, are progressively updated during training to retain historical source and target global representations. New features are incrementally enqueued while the oldest are discarded upon reaching the capacity n, thereby preserving a temporally balanced representation that evolves smoothly without being skewed by recent updates. A multiscale Euclidean-MMD metric facilitates the adaptive selection of features near g_m^t. Euclidean distances between g_m^t and features in M_{sg} and M_{tg} identify the nearest intradomain candidate sets $\widetilde{M}_{sg}$ and $\widetilde{M}_{tg}$, each with n_{top} candidates. MMD then measures the distribution distances D_{g1} between g_m^t and $\widetilde{M}_{sg}$, and D_{g2} between g_m^t and $\widetilde{M}_{tg}$. D_{g1} is derived via the squared form of MMD [16] as follows:

$$D_{g1} = \frac{1}{(n_{top})^2} \sum_{k=1}^{n_{top}} \sum_{l=1}^{n_{top}} r\left(\tilde{g}_k^s, \tilde{g}_l^s\right) + 1 - \frac{2}{n_{top}} \sum_{k=1}^{n_{top}} r\left(g_m^t, \tilde{g}_k^s\right) \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

where $r(\cdot,\cdot)$ is the gaussian kernel in the reproducing Hilbert space, defined as $r(\tilde{g}_k^s, \tilde{g}_l^s) = \exp(-|\tilde{g}_k^s - \tilde{g}_l^s|^2/2\vartheta^2)$, with $\vartheta = 1$. The value 1 in (5) follows from $r(g_m^t, g_m^t) = 1$. D_{g2} can be achieved by substituting $\tilde{g}_k^s$ and $\tilde{g}_l^s$ in (5) with $\tilde{g}_k^t$ and $\tilde{g}_l^t$. Close-proximity features typically share the same category [36], but domain bias may render D_{g1} and D_{g2} comparison misleading for source candidates selection. Thus, a holistic MMD distance D_{g3} is computed between M_s and M_t to capture cross-domain differences

$$D_{g3} = \frac{1}{n^2}(R_{ss} + R_{tt} - 2R_{st}) \tag{6}$$

where $R_{ss} = \sum_{k=1}^{n}\sum_{l=1}^{n} r(\tilde{g}_k^s, \tilde{g}_l^s)$, $R_{tt} = \sum_{k=1}^{n}\sum_{l=1}^{n} r(\tilde{g}_k^t, \tilde{g}_l^t)$, $R_{st} = \sum_{k=1}^{n}\sum_{l=1}^{n} r(\tilde{g}_k^s, \tilde{g}_l^t)$. Then, the refined probability $\tilde{p}$ is obtained by selecting neighbors via D_{g1}, D_{g2}, and D_{g3}

$$\tilde{p} = \begin{cases} \mathbb{E}_{\tilde{g}_k^s \in \widetilde{M}_{sg}} \mathrm{sm}\left(F\left(\tilde{g}_k^s\right)\right), & D_{g1} + D_{g3} < D_{g2} \\ \mathbb{E}_{\tilde{g}_k^t \in \widetilde{M}_{tg}} \mathrm{sm}\left(F\left(\tilde{g}_k^t\right)\right), & D_{g1} + D_{g3} \geq D_{g2} \end{cases} \tag{7}$$

where $\mathbb{E}$ denotes the expectation calculation. If $D_{g1} + D_{g3} < D_{g2}$, indicating that $\widetilde{M}_s$ is closest to g_m^t after mitigating domain bias, $\widetilde{M}_s$ and g_m^t likely share the same class label. In this case, the refined probability is computed from the expectation of $\widetilde{M}_s$'s high-confidence predictions; otherwise, the average probability of $\widetilde{M}_t$ is used. The refined label $\breve{y}_m = \mathrm{argmax}(\tilde{p})$ is then obtained, and its reliability $\breve{\lambda}_m$ is calculated using the negative exponential of the information entropy [37], $\breve{\lambda}_m = \exp(-\mathcal{H}(\tilde{p})/\mathcal{H}_{max})$. Refining each g_m^t in G_t produces a global pseudolabel set $\breve{Y} = \{\breve{y}_m\}_{m=1}^{n_t}$ with confidence values $\breve{\Lambda} = \{\breve{\lambda}_m\}_{m=1}^{n_t}$, and similarly, the local pseudolabel set $\hat{Y}$ and confidence values $\hat{\Lambda}$ are derived.

It is worth noting that ADFS is inherently robust to challenging adaptation scenarios, including large source–target distribution discrepancies and severe class imbalance. Taking the pseudolabel refinement of a target-domain feature g_m^t as an example, the holistic MMD term D_{g3} captures cross-domain distribution differences; when the gap is large, an increased D_{g3} makes the condition $D_{g1} + D_{g3} < D_{g2}$ harder to satisfy, thereby discouraging reliance on potentially biased source neighbors and alleviating negative transfer. For extremely rare categories, insufficient supportive neighbors result in higher uncertainty in the refined probability distribution, which is reflected by the entropy-aware confidence score $\breve{\lambda}_m$ and consequently down-weights unreliable pseudolabels. Through adaptive memory selection and confidence-aware weighting, ADFS prioritizes pseudolabel reliability over quantity, enabling stable refinement under both severe domain shift and class imbalance.

C. Prototype Contrastive Learning via SATS

Existing contrastive learning methods suffer from prototype drift and intraclass dispersion under domain shifts [28], [38], hampering effective alignment. To address these challenges, we introduce a novel SATS-driven contrastive approach that leverages source supervision to calibrate class prototypes, fostering more discriminative and compact representations at both global and local feature levels. Specifically, SGLPCL constructs and updates global and local prototypes from global features (G_s, G_t) and local features (Z_s, Z_t), respectively. It then simultaneously performs temperature-calibrated prototype contrastive learning in both the global and local branches, thereby forming a unified global–local prototype contrastive learning framework. Since the two branches follow an almost identical computational pipeline, differing only in the input feature type (global or local features), we detail the global branch below to avoid redundancy.

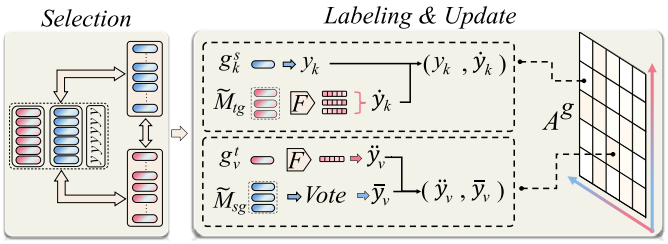


Fig. 3. ATS matrix construction diagram.

First, the source and target prototype update momentum are derived from G_s, G_t, and $\breve{Y}$

$$\Delta\mu_k^t = \mathbb{E}_{(g_m^t, \breve{y}_m)\sim(G_t, \breve{Y})}\left\{g_m^t \mid \breve{y}_m = k\right\} \tag{8}$$

where $\Delta\mu_k^t$ represent the target momentum for the kth class. $\Delta\mu_k^s$ can be computed by replacing G_t and $\breve{Y}$ with G_s and Y in (8). Next, source and target prototypes are derived and updated via a moving average

$$\mu_k^s = \beta \cdot \mu_k^{s(*)} + (1-\beta)\cdot\Delta\mu_k^s \tag{9}$$

where μ_k^s denotes the kth class prototype in the source domain, β is the momentum factor, and $\mu_k^{s(*)}$ is the previous prototype. The target prototype μ_k^t is calculated as follows:

$$\mu_k^t = (1-\beta)\cdot\frac{\mu_k^{s(*)}}{\|\mu_k^{s(*)}\|_2} + \beta\cdot\frac{\Delta\mu_k^t}{\|\Delta\mu_k^t\|_2} \tag{10}$$

where $\|\cdot\|_2$ denotes the L2 norm. By normalizing source prototypes and target momentum in a unified space, target prototypes align with source centers, reducing prototype shift from erroneous pseudolabels. Cosine similarity between source prototypes $\mu^s = \{\mu_k^s\}_{k=1}^c$ and target prototypes $\mu^t = \{\mu_k^t\}_{k=1}^c$ yields the global similarity matrix $S^g = \mu^s(\mu^t)^{\mathrm{T}}$, a $c \times c$ matrix where S_{uv}^g represents similarity between μ_u^s and target prototype μ_v^t. Calibration is then achieved with an ATS matrix A^g. To this end, similar to feature selection in pseudolabel refinement, g_k^s and g_v^t are selected based on proximity to the target and source memory pools, respectively. The next step in constructing A^g is illustrated in Fig. 3.

From Fig. 3, cross-domain label pairs $(y_k, \dot{y}_k)$ and $(\ddot{y}_v, \bar{y}_v)$ are formed by combining g_k^s with target memory neighbors $\widetilde{M}_{tg}$ and g_v^t with source memory neighbors $\widetilde{M}_{sg}$. Specifically, akin to (7), the refined label $\dot{y}_k$ is derived by optimizing the average probability $\tilde{p}$ from $\widetilde{M}_{tg}$. $\dot{y}_k$ and the label y_k from g_k^s comprise the first label pair $(y_k, \dot{y}_k)$. For the second pair, $\ddot{y}_v$ is derived from g_v^t, and the refined label $\bar{y}_v$ is determined from the highest vote count in $\widetilde{M}_{sg}$

$$\bar{y}_v = \underset{k\in C}{\mathrm{argmax}} \sum_{u=1}^{n_{top}} \mathbf{1}(y_u = k) \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

where C denotes the category index set $\{1, \ldots, c\}$, y_u is the source label of $\tilde{g}_u^s$ in $\widetilde{M}_{sg}$, and $\mathbf{1}(\cdot)$ is the indicator function that equals 1 if $y_u = k$ and 0 otherwise. Note that both label pair types contain multiple pairs in practice. A_g, a $c \times c$ matrix initialized to zero, is updated based on the cross-domain label pairs: $A^g_{y_k,\dot{y}_k} \leftarrow A^g_{y_k,\dot{y}_k} + 1$ and $A^g_{\ddot{y}_v,\bar{y}_v} \leftarrow A^g_{\ddot{y}_v,\bar{y}_v} + 1$. These label pairs assume similar samples in feature space belong to the same category, such that a higher A^g_{uv} indicates greater overlap between source class u and target class v, reflecting lower intraclass compactness. Thus, S^g requires calibration to adjust the penalty based on A^g. To this end, A^g is first normalized along the source (src) and target (tar) dimension, yielding $\breve{A}^g = \mathrm{sm}_{src}(A^g)$ and $\hat{A}^g = \mathrm{sm}_{tar}(A^g)$. Then, S^g is statistically calibrated

$$\tilde{S}^g = \frac{1}{2}\left(S^g \odot \left(\breve{A}^g + \rho\right) + S^g \odot \left(\hat{A}^g + \rho\right)\right) \tag{12}$$

where $\tilde{S}^g$ is the calibrated similarity matrix, $\odot$ denotes element-wise multiplication, and ρ is a fixed offset ($1e^{-6}$). Replacing global with local features yields calibrated local similarity $\tilde{S}^l$. Finally, the total prototype contrastive loss $\mathcal{L}_{ctr}$ is defined using CE [34]

$$\mathcal{L}_{ctr} = \mathrm{CE}\left(\tilde{S}^g, I\right) + \mathrm{CE}\left(\tilde{S}^l, I\right) \tag{13}$$

where I is a $c \times c$ binary matrix with 1 on the diagonal for positive pairs from the same class and 0 off-diagonal for negative pairs from different classes.

D. Joint Optimization

Beyond the contrastive loss introduced in Sections II-A–II-C to drive cross-domain alignment, we jointly incorporate a classification loss to promote more stable and target-aware adaptation. This classification loss $\mathcal{L}_{cls}$ comprises two complementary components, with one defined on the labeled source domain and the other formulated for the unlabeled target domain. For the labeled source-domain loss $\mathcal{L}^s_{cls}$

$$\mathcal{L}^s_{cls} = \mathbb{E}_{g^s_m \in G_s}\mathrm{CE}\left(g^s_m, y_m\right) + \mathbb{E}_{z^s_m \in Z_s}\mathrm{CE}\left(z^s_m, y_m\right). \tag{14}$$

The unlabeled target-domain loss is computed via negative learning [39] and unbiased confidence estimation

$$\begin{aligned}\mathcal{L}^t_{cls} = {} & \mathbb{E}_{g^t_m \in G_t}\left[-\breve{\lambda}_m \cdot \sum_{k=1}^{c} \mathrm{O}\left(\breve{y}'_m\right)_k \log\left(1 - p^g_k\right)\right] \\ & + \mathbb{E}_{z^t_m \in Z_t}\left[-\hat{\lambda}_m \sum_{k=1}^{c} \mathrm{O}\left(\hat{y}'_m\right)_k \cdot \log\left(1 - p^l_k\right)\right]\end{aligned} \tag{15}$$

where $\breve{\lambda}_m$ is the confidence for the refined pseudolabel $\breve{y}_m$, and $\breve{y}'_m$ is a complementary label from $C \setminus \breve{y}_m$ with its one-hot vector $\mathrm{O}(\breve{y}'_m)$. $p^g = \mathrm{sm}(F(g^t_m))$ represents the global feature-based probability distribution. Notation for local loss aligns with that of global loss. Thus, the total classification loss is $\mathcal{L}_{cls} = \mathcal{L}^s_{cls} + \mathcal{L}^t_{cls}$. The network parameters θ are optimized through classification and contrastive losses

$$\min_{\theta}\left(\mathcal{L}_{cls} + \psi\mathcal{L}_{ctr}\right) \tag{16}$$

where ψ is tradeoff coefficient. The pseudocode for the SGLPCL procedure is presented in Algorithm 1.

Algorithm 1 Pseudocode Implementation of SGLPCL

Input: Source-domain samples X_s with corresponding labels Y, target-domain samples X_t, and training iterations N.
Output: Trained network parameters θ.

```
Randomly initialize the network parameters θ.
for i = 1 to N do
    Z_s, Z_t ← E(X_s), E(X_t)
    Obtain global features G_s and G_t via Eqs. (1)-(4).
    Refine pseudolabels Y̆ and Ŷ via Eqs. (5)–(7).
    Compute the contrastive loss L_ctr based on (G_s, Y, G_t, Y̆) and (Z_s, Y, Z_t, Ŷ) using Eqs. (8)-(13).
    Compute the classification loss L_cls based on (G_s, Y, G_t, Y̆) and (Z_s, Y, Z_t, Ŷ) using Eqs. (14)-(15).
    Update θ by minimizing Eq. (16).
end for
```

III. Experiments and Analysis

A. Dataset Description

We evaluate the proposed algorithm on three hyperspectral cross-domain datasets: Houston, HyRANK, and Shanghai–Hangzhou (S–H). The Houston dataset consists of the source-domain scene, Houston2013 [40], with 349×1905 pixels and 144 spectral bands, and the target-domain scene Houston2018 [41], with 48 spectral bands. Both scenes cover seven classes. For analysis, 48 spectral bands are extracted from Houston2013, focusing on a 210×954 pixel overlapping area. The HyRANK dataset [42] includes Dioni as the source domain and Loukia as the target, with sizes of 250×1376 and 249×945 pixels, respectively, and contains 176 spectral bands and 12 classes. The S–H dataset includes the source scene Hangzhou of size 590×230 pixels and the target scene Shanghai of size 1600×230 pixels, with 198 spectral bands and three shared classes. class distributions are shown in Table I.

B. Experimental Settings

1) Compared Methods: To comprehensively evaluate the effectiveness and generalization capability of SGLPCL, we compare it with a broad range of state-of-the-art HSDA methods. Specifically, we benchmark against two representative UDA baselines, joint adversarial domain adaptation (JADA) [11] and bi-classifier determinacy maximization (BCDM) [10], along with eight advanced HSDA approaches covering different adaptation paradigms. These methods are categorized into four groups: 1) distance metric learning methods, represented by topological structure and semantic information transfer network (TSTnet) [16], which learn transferable distance metrics for domain alignment; (2) adversarial learning methods, including unsupervised JADA (UJADA) [20], joint adversarial network with semantic and topology fusion (JAN-STF) [17], and category semantic guided UDA network (CSGNet) [18], which employ domain discriminators or semantic adversarial objectives to reduce cross-domain distribution discrepancies; and (3) self-training methods, such as bi-classifier adversarial network (BCAN) [25] and pseudolabel-assisted subdomain adaptation (PASDA) [24], which iteratively refine pseudolabels to strengthen target-domain supervision; 4) contrastive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SAMPLE DISTRIBUTION ACROSS THREE DATASETS

Houston (Houston2013 → Houston2018)			
No.	Class	Source	Target
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total samples		2530	53200
HyRANK (Dioni → Loukia)			
No.	Class	Source	Target
1	Dense urban fabric	1262	206
2	Mineral extraction sites	204	54
3	Non irrigated arable land	614	426
4	Fruit trees	150	79
5	Olive groves	1768	1107
6	Coniferous forest	361	422
7	Dense sderophyllous vegetation	5035	2996
8	Sparce sderophyllous vegetation	6374	2361
9	Sparcely vegetated areas	1754	399
10	Rocks and sand	492	453
11	Water	1612	1393
12	Coastal water	398	421
Total samples		20024	10317
S-H (Hangzhou → Shanghai)			
No.	Class	Source	Target
1	Water	18043	123123
2	Land/Building	77450	161689
3	Plant	40207	83188
Total samples		135700	368000

learning methods, represented by supervised contrastive learning-based UDA (SCLUDA) [28] and multilevel UDA (MLUDA) [30], which achieve discriminative feature alignment via contrastive optimization.

2) Unified Configuration: All experiments are conducted in PyTorch [43] on an NVIDIA 3080 Ti GPU. Following [16], we randomly select a small portion (5% by default) of the source-domain data and an equal amount of unlabeled target-domain data for training, with evaluation on the entire target domain. Model parameters are optimized using Adam [44] for 150 epochs, with an initial learning rate ∂_0 of 1×10^{-3} and an adaptive decay schedule $\partial = \partial_0/(1+10{\cdot}\delta)^{3/4}$, where δ increases linearly during training. To ensure fair comparison, all methods employ VGG16 [45] as the backbone for feature extraction with 12×12 input patches, and results are averaged over ten trials. To ensure a fair comparison across all methods, we adopt VGG16 [45] as the default backbone for feature extraction. Despite being relatively lightweight, VGG16 remains a widely used and well-established architecture in HSDA research [10], [16], offering stable feature representations and efficient convergence, while avoiding potential domain-specific bias that deeper or transformer-based networks may introduce in small-sample HSDA settings. Results are averaged over ten trials for statistical reliability. Classification performance is evaluated using overall accuracy (OA), average accuracy (AA), and Cohen's Kappa coefficient (KC) [46], which, respectively, measure overall correctness, class-balanced accuracy, and prediction consistency beyond chance. Higher values of all three metrics indicate better performance.

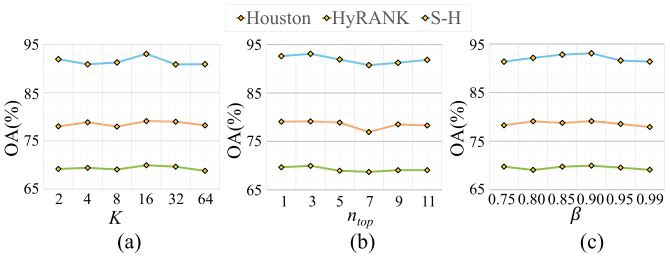


Fig. 4. OA (%) performance for different values of (a) K, (b) n_{top}, and (c) β across the Houston, HyRANK, and S–H datasets.

C. Parameters Analysis and Settings

1) Key Parameters: The SGLPCL algorithm contains several critical parameters, among which three directly govern its internal training mechanism: the number of multihead attention mechanisms (K), the capacity of the candidate feature set (n_{top}), and the momentum factor (β) that controls the update rate of class prototypes. To determine suitable configurations, we conduct control-variable experiments where K, n_{top}, and β are varied within the ranges of [2, 4, 8, 16, 32, and 64], [1, 3, 5, 7, 9, and 11], and [0.75, 0.80, 0.85, 0.90, 0.95, and 0.999], respectively, each evaluated over ten trials. Fig. 4 presents the average OA results corresponding to different parameter choices.

As shown in Fig. 4, SGLPCL exhibits strong robustness across all three hyperparameters, with performance remaining consistently stable within a broad range. This robustness stems from the model's mutually guided learning mechanism, where the source-guided prototype construction and ADFS jointly regularize the representation space. Consequently, minor variations in K, n_{top}, or β have a limited effect on optimization dynamics. The optimal configuration (K, n_{top}, and β) is found to be (16, 3, and 0.90) across all datasets, balancing discriminative capacity, convergence stability, and computational efficiency. These values are thus adopted as the default hyperparameter setting in subsequent experiments. Furthermore, the memory pool size n is empirically set to 3000 to ensure a balance between computational cost and representation diversity. Preliminary observations show that smaller pools lead to unstable prototype updates, while larger ones yield marginal gains at higher cost, consistent with prior memory-based contrastive frameworks [47], [48].

In addition, three parameter groups influence the overall training behavior of SGLPCL: training batch size, resolution of input patches (patch size), and tradeoff coefficient ψ in the overall training objective of (16). Similarly, we investigate the effects of batch size, patch size, and coefficient ψ by varying them over the ranges [16, 32, 64, 128, and 256], [10, 12, 14, 16, and 18], and [0.25, 0.5, 1, 2, and 4], respectively. Note that a patch size of 12 refers to an input patch with a resolution of 12×12. The experimental results are illustrated in Fig. 5.

The batch size directly determines the coverage of the graph construction in GAT, with larger batches allowing broader node interactions within a single batch. However, as shown in Fig. 5(a), SGLPCL demonstrates strong stability across different batch sizes on all three datasets. This robustness stems

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

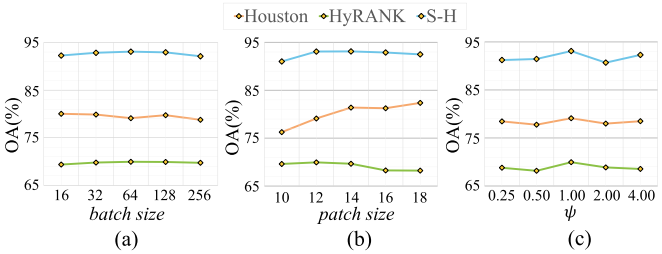


Fig. 5. OA (%) performance for different values of (a) batch size, (b) patch size, and (c) ψ across the Houston, HyRANK, and S–H datasets.

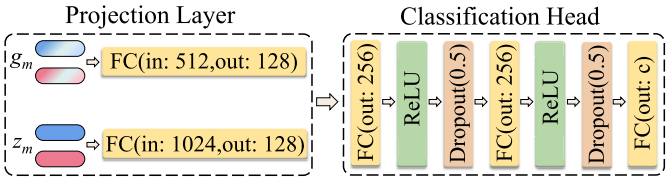


Fig. 6. Structure of the classifier.

from GAT's multiround, multibatch random sampling strategy: as the number of iterations increases, the impact of the sampling range within each batch is gradually smoothed, allowing each node to interact comprehensively with all nodes in the domain, thereby learning stable and holistic global features. As a result, GAT exhibits minimal dependency on batch size. Considering the slightly better performance on the HyRANK and S–H datasets, we set a batch size of 64 as the default across all datasets. The patch size affects the composition of each node's raw local features during GAT graph construction, with this influence closely tied to the class distribution of the dataset. As shown in Fig. 5(b), increasing the patch size on the Houston dataset enables the incorporation of richer contextual information, leading to notable performance gains. In contrast, for the HyRANK dataset, which contains a larger number of classes with more dispersed distributions, larger patches tend to include excessive information from interfering categories, inducing negative learning and causing a slight performance degradation. For the S–H dataset, characterized by fewer and more compact classes, performance remains relatively stable across different patch sizes. Considering the results across all three datasets and the settings used in prior work [16], we adopt 12×12 as the unified default configuration. Fig. 5(c) shows that varying ψ yields consistent performance trends across all three datasets, indicating that the loss formulation in (16) generalizes well across datasets. Since $\psi = 1$ achieves the best performance in all cases, it is adopted as the unified default weight for the contrastive loss.

2) Network Parameters: The network architecture of the proposed SGLPCL algorithm primarily consists of two components: a feature extractor and a classifier. The feature extractor employs VGG16, a well-established model widely used in HSIC tasks, as detailed in [45]. The classifier is designed to process the extracted features and output class labels. The detailed structure of the classifier is shown in Fig. 6.

As illustrated in Fig. 6, the classifier consists of a projection layer and a classification head. The projection layer maps the global feature $g_m \in \mathbb{R}^{512}$ and the local feature $z_m \in \mathbb{R}^{1024}$ into a unified 128-D embedding through fully connected (FC) layers. This dimensionality compression not only aligns heterogeneous feature spaces but also facilitates prototype-level contrastive learning by maintaining metric consistency and reducing feature redundancy. The classification head further processes the 128-D embedding via a series of FC layers with ReLU activation and dropout (rate = 0.5), introducing nonlinearity and improving generalization. The use of dropout helps prevent overfitting by randomly deactivating neurons during training, enhancing model robustness. Overall, this design strikes a balance between representational richness, robustness, and computational efficiency, providing a solid foundation for global–local prototype learning.

D. Classification Results and Analysis

1) Quantitative Analysis: To evaluate transfer performance with limited labeled data, we begin by analyzing results under the default 5% source sampling ratio. Tables II–IV report the quantitative classification results of the proposed SGLPCL and ten leading HSDA methods across three benchmark datasets. All results are averaged over ten independent runs to ensure statistical reliability. Moreover, we further conduct paired two-sided Student's t-tests [49], where p-values (p) are computed and compared against a significance threshold of 0.05, to assess whether the observed improvements are statistically significant or within the expected run-to-run variance. As shown in Table II, SGLPCL achieves the highest overall performance on the Houston dataset, outperforming the second-best method by 0.26%, 2.06%, and 4.37% in AA, OA, and KC. Although the AA improvement is marginal and statistically insignificant ($p > 0.05$), falling within expected run-to-run variance, SGLPCL attains statistically significant gains in both OA ($p = 0.039$) and KC ($p = 0.003$), indicating more reliable global classification and interclass agreement. These improvements are primarily attributed to the proposed global–local category prototype consistency learning, which yields optimal or near-optimal performance on most classes (classes 2, 3, 5, and 6). A closer inspection at the class level reveals relatively weaker performance on a few categories (classes 1, 4, and 7). For example, the degradation of Grass healthy (class 1) is mainly due to its high spectral similarity to Grass stressed, which exacerbates residual domain discrepancies. For Water (class 4), the extremely limited number of target-domain samples (only 22 samples) hinders reliable prototype estimation and leads to unstable predictions. For the HyRANK benchmark, which exhibits significant domain shifts and pronounced class imbalance due to geographical differences, the experimental results in Table III demonstrate that SGLPCL still consistently outperforms all comparison methods. Despite a relatively small and statistically insignificant improvement in AA ($p > 0.05$), which can be attributed to class imbalance and the inherent bias of the macro average toward underrepresented classes, SGLPCL achieves notable and statistically significant gains of 1.96% ($p = 0.027$) in OA and 2.07% ($p = 0.019$) in KC. This indicates that while AA may not fully capture performance in imbalanced settings, SGLPCL excels in enhancing global accuracy and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CLASSIFICATION ACCURACIES OF 11 METHODS ON TARGET-DOMAIN (HOUSTON2018) USING 5% OF SOURCE-DOMAIN (HOUSTON2013) SAMPLES

Class	JADA	BCDM	TSTnet	UJADA	BCAN	PASDA	SCLUDA	JAN-STF	MLUDA	CSGNet	SGLPCL
1	79.34	93.11	84.01	90.03	69.56	01.09	64.86	83.75	72.09	45.14	77.89
2	71.74	60.12	67.56	68.90	82.82	81.31	68.53	75.10	71.27	86.51	86.90
3	66.29	62.72	53.12	63.80	57.40	24.92	57.98	67.74	54.03	55.16	66.97
4	95.45	94.55	47.73	99.09	63.18	04.09	20.91	98.18	05.91	20.00	90.00
5	75.72	74.53	69.77	81.95	80.49	75.16	81.46	78.38	84.35	72.64	81.16
6	74.71	65.98	74.14	67.54	76.07	81.73	84.70	77.04	84.60	82.73	82.08
7	61.50	70.08	57.89	66.93	53.14	61.70	51.85	64.73	48.01	52.21	61.78
AA	74.97 ±2.98	74.44 ±2.13	64.89 ±3.30	76.89 ±1.75	68.95 ±3.13	47.14 ±3.39	61.47 ±4.29	77.85 ±1.02	60.15 ±3.82	59.20 ±4.00	**78.11** ±2.96
OA	72.65 ±3.03	67.33 ±4.73	70.30 ±3.29	69.43 ±3.85	73.25 ±3.03	73.60 ±1.35	77.04 ±1.74	75.22 ±1.55	77.05 ±1.91	75.99 ±1.49	**79.11** ±1.45
KC	58.36 ±3.07	52.83 ±2.56	54.51 ±3.97	55.39 ±4.36	58.16 ±3.85	57.64 ±2.92	62.43 ±2.80	61.80 ±1.76	62.38 ±2.77	61.10 ±2.44	**66.80** ±2.17

TABLE III
CLASSIFICATION ACCURACIES OF 11 METHODS ON TARGET-DOMAIN (LOUKIA) USING 5% OF SOURCE-DOMAIN (DIONI) SAMPLES

Class	JADA	BCDM	TSTnet	UJADA	BCAN	PASDA	SCLUDA	JAN-STF	MLUDA	CSGNet	SGLPCL
1	30.92	40.87	31.21	17.14	10.25	23.30	13.30	29.76	04.22	34.22	18.54
2	65.93	41.85	28.33	30.37	31.71	32.22	10.37	56.85	06.30	66.48	56.67
3	34.39	36.36	30.63	07.65	10.21	25.99	00.00	40.31	05.26	41.95	51.92
4	25.06	21.65	11.90	01.01	08.54	01.52	00.00	20.76	03.67	17.97	26.33
5	64.98	59.24	65.32	56.87	20.34	72.81	05.43	73.62	17.68	05.55	69.12
6	14.83	25.59	04.08	11.42	21.95	23.01	01.42	19.31	12.58	36.47	15.71
7	69.22	67.16	73.17	67.76	85.32	59.21	84.29	68.26	80.70	71.60	75.66
8	70.27	66.38	66.93	66.75	71.23	71.54	67.07	70.84	69.09	67.54	72.63
9	73.33	75.54	78.97	77.09	56.02	38.57	37.57	55.51	30.38	58.95	57.84
10	05.89	00.13	05.17	30.25	00.00	14.72	06.23	19.38	12.03	00.53	10.53
11	100.0	100.0	99.93	100.0	100.0	100.0	100.0	99.83	100.0	100.0	100.0
12	100.0	99.74	100.0	100.0	97.89	40.00	99.14	99.95	93.87	100.0	100.0
AA	54.57 ±3.86	52.88 ±4.50	49.64 ±2.87	47.20 ±3.06	42.79 ±3.04	41.91 ±7.22	35.40 ±4.01	54.53 ±4.82	36.31 ±2.83	50.10 ±2.34	**54.58** ±3.69
OA	67.01 ±1.13	65.30 ±1.31	66.72 ±1.43	64.20 ±1.21	64.68 ±1.71	61.31 ±1.80	60.06 ±1.42	67.99 ±1.79	61.05 ±0.93	61.12 ±0.87	**69.95** ±1.60
KC	59.82 ±1.42	58.03 ±1.72	59.66 ±1.76	56.37 ±1.57	55.84 ±2.38	52.51 ±2.36	48.67 ±2.37	61.15 ±2.03	50.55 ±1.51	51.81 ±1.15	**63.22** ±1.99

TABLE IV
CLASSIFICATION ACCURACIES OF 11 METHODS ON TARGET-DOMAIN (SHANGHAI) USING 5% OF SOURCE-DOMAIN (HANGZHOU) SAMPLES

Class	JADA	BCDM	TSTnet	UJADA	BCAN	PASDA	SCLUDA	JAN-STF	MLUDA	CSGNet	SGLPCL
1	84.51	90.41	89.00	93.24	92.60	91.36	92.52	89.37	92.09	96.27	94.01
2	85.64	73.73	82.75	78.64	82.33	89.18	89.42	89.04	89.12	84.62	90.24
3	88.59	99.93	96.68	99.70	98.99	94.24	96.23	92.49	94.51	93.02	97.27
AA	86.25 ±5.27	88.02 ±1.53	89.48 ±1.86	90.53 ±1.11	91.31 ±2.80	91.58 ±3.26	92.72 ±0.51	90.30 ±3.26	91.91 ±1.81	91.30 ±2.31	**93.84** ±0.85
OA	85.93 ±3.48	85.23 ±1.93	88.79 ±2.68	88.28 ±1.42	89.54 ±3.53	91.05 ±1.27	92.00 ±0.94	89.93 ±2.34	91.33 ±1.52	90.42 ±3.10	**93.09** ±1.27
KC	78.16 ±5.75	77.81 ±2.80	81.69 ±3.81	82.29 ±2.08	84.21 ±5.23	86.20 ±1.84	87.68 ±1.40	84.40 ±3.77	86.63 ±2.37	85.38 ±4.49	**89.38** ±1.88

class separation, as reflected by OA and KC. Such superiority is largely attributed to the ADFS-based pseudolabel refinement strategy, which enhances pseudolabel reliability under challenging adaptation conditions. Notably, although BCAN and PASDA also rely on pseudolabel-based self-training, SGLPCL achieves substantial OA improvements of 5.27% and 8.58%, respectively, highlighting the advantage of ADFS in refining pseudolabels rather than merely exploiting them. Moreover, SGLPCL attains a 1.27% higher classification accuracy on the extremely rare class Fruit trees (the fourth class) than the second-best method. These results verify that the proposed ADFS strategy effectively handles challenges induced by large domain discrepancies and mitigates the performance degradation of extremely rare classes caused by severe class imbalance. On the S–H dataset (Table IV), the proposed SGLPCL demonstrates superior generalization capability, statistically significantly outperforming the second-best method (SCLUDA) by 1.12% (p = 0.011), 1.09% (p = 0.049), and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

TABLE V
OA(%) OF NINE METHODS ACROSS VARYING RATIOS OF TRAINING SAMPLES ON THE HOUSTON DATASET

Ratios	T	U	B	P	S	J	M	C	Ours
10%	74.3	75.6	77.2	73.5	78.1	78.2	77.2	77.0	**81.5**
20%	76.8	76.2	79.1	74.8	79.0	79.2	79.0	79.5	**81.8**
40%	78.6	77.5	78.7	75.0	79.6	79.5	78.3	80.9	**82.7**
80%	78.3	77.6	80.1	76.0	79.3	80.2	79.6	81.4	**82.9**
100%	79.4	78.1	80.8	76.6	80.6	80.4	81.4	81.6	**83.0**

1.70% (p = 0.046) in AA, OA, and KC, respectively. These improvements in overall metrics are primarily reflected at the class level by optimal or near-optimal performance on classes 1 and 2, arising from the synergy between source-guided supervision and global–local prototype contrastive learning, which enhances intraclass compactness and interclass separation. The relatively less competitive performance on Plant (class 3) is mainly attributable to vegetation diversity and phenological differences that induce subtle spectral shifts under severe cross-scene domain discrepancy. Overall, SGLPCL consistently achieves the best performance and convergence stability across all benchmarks. Statistical tests indicate that while AA improvements in certain datasets fall within the expected run-to-run variance, the gains in OA and KC are statistically significant and clearly separated from competing methods, reinforcing the robustness of SGLPCL for HSDA.

To further assess the robustness of different source sampling ratios, we compare SGLPCL with eight representative HSDA methods—TSTnet (T), UJADA (U), BCAN (B), PASDA (P), SCLUDA (S), JAN-STF (J), MLUDA (M), and CSGNet (C). Large-scale experiments are first conducted on the Houston dataset, evaluating OA performance across sampling ratios ranging from 10% to 100%, as summarized in Table V.

According to Table V, SGLPCL consistently surpasses the other eight HSDA methods on the Houston dataset across different source sampling ratios, achieving improvements of 1.61%–3.31% over the second-best methods. At 10% sampling, it achieves 81.5% OA, surpassing SCLUDA by 3.4%, demonstrating strong transfer capability under minimal source supervision. While most methods improve as the sampling ratio increases, some exhibit instability—for example, SCLUDA's OA drops by 0.33% from 40% to 80% due to pseudolabel noise. In contrast, SGLPCL maintains stable growth, rising from 82.7% at 40% to 83.0% at full supervision. This robustness stems from the integration of source-driven supervision with multilevel global–local prototype alignment, which mitigates error propagation and ensures consistent interclass separability. Moreover, small-scale experiments on the HyRANK and S–H datasets are conducted with source sampling ratios from 1% to 10%, and the results are summarized in Fig. 7.

As illustrated in Fig. 7, performance generally improves with higher source sampling ratios. On the HyRANK dataset [Fig. 7(a)], graph-based methods such as TSTnet and JAN-STF outperform traditional contrastive and self-training approaches by more than 2%, highlighting the advantage of neighborhood aggregation for spatial coherence. Across all ratios, SGLPCL

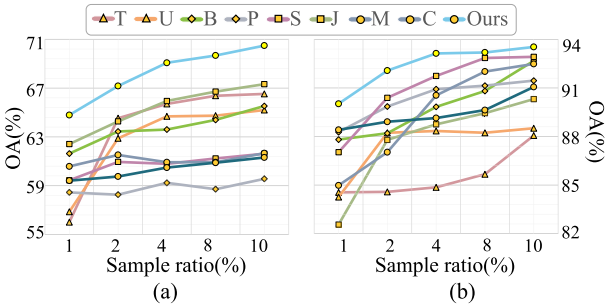


Fig. 7. OA results of nine methods across varying ratios of training samples for the two dataset. (a) HyRANK. (b) S–H.

consistently surpasses all competitors by 2.38%–3.16%, benefiting from its integrated spectral–spatial–semantic prototype learning, which enhances category separability and mitigates domain noise. The method's ability to simultaneously learn both global and local features makes it particularly effective for cross-domain tasks. On the S–H dataset [Fig. 7(b)], SCLUDA performs robustly from 2% sampling forward due to instance-level contrastive learning. In contrast, SGLPCL leverages source-domain supervision and global–local prototype alignment to further improve intraclass compactness and interclass discrimination. It outperforms the next-best method by 0.34%–1.75% across all sampling ratios, demonstrating robust transfer under limited supervision.

To validate the generality of the proposed method across different backbones, we replace VGG16 with the widely adopted spectral–spatial backbone deep residual 3DCNN [50] and reevaluate all methods under identical settings for fair comparison. The results are summarized in Table VI.

As shown in Table VI, SGLPCL consistently outperforms all competing methods when adopting a deep residual 3DCNN as the backbone. In terms of the primary OA metric, SGLPCL achieves clear and statistically significant improvements over the second-best method on all three datasets, with gains of 2.22% (p = 0.022), 1.08% (p = 0.001), and 2.30% (p = 0.047), respectively. A further comparison between VGG16 and deep residual 3DCNN backbones reveals consistent performance trends across datasets. On Houston and HyRANK, which have more classes, most competing methods suffer OA degradation when adopting deep residual 3DCNN (e.g., −2.04% and −2.65% for CSGNet and SCLUDA on Houston, and −3.36% and −3.20% for TSTnet and BCAN on HyRANK). A plausible explanation is that, under limited labeled training data (5%) and complex class distributions, patch-based spectral–spatial modeling may introduce redundant or noisy spectral information, thereby adversely affecting the discriminative capability of certain spectral feature extractors. Despite this, SGLPCL consistently maintains the best performance, demonstrating strong robustness to backbone variations. Conversely, on the S–H dataset with fewer classes and lower spectral redundancy, deep residual 3DCNN generally improves most methods over VGG16, for example, yielding OA gains of 1.58%, 1.95%, and 1.11% for SGLPCL, CSGNet, and JAN-STF. These results confirm that SGLPCL generalizes effectively across different backbone architectures while preserving stable and competitive performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CLASSIFICATION RESULTS OF 11 METHODS ON THREE DATASETS USING 5% OF SOURCE-DOMAIN SAMPLES FOR TRAINING

Methods	Houston (Houston2013→Houston2018)			HyRANK (Dioni→Loukia)			S-H (Hangzhou→Shanghai)		
	AA(%)	OA(%)	KC(%)	AA(%)	OA(%)	KC(%)	AA(%)	OA(%)	KC(%)
JADA	$56.85_{\pm 4.2}$	$71.84_{\pm 4.4}$	$54.92_{\pm 4.3}$	$47.74_{\pm 3.8}$	$61.65_{\pm 2.5}$	$52.67_{\pm 3.3}$	$86.76_{\pm 5.7}$	$86.59_{\pm 5.4}$	$79.32_{\pm 8.2}$
BCDM	$62.52_{\pm 2.1}$	$65.68_{\pm 4.1}$	$50.07_{\pm 3.5}$	$46.32_{\pm 2.1}$	$62.73_{\pm 1.5}$	$54.83_{\pm 1.7}$	$89.91_{\pm 3.5}$	$87.75_{\pm 4.5}$	$81.45_{\pm 6.5}$
TSTnet	$59.03_{\pm 4.7}$	$74.64_{\pm 1.7}$	$58.79_{\pm 3.6}$	$45.50_{\pm 2.5}$	$63.36_{\pm 0.8}$	$54.97_{\pm 1.3}$	$90.13_{\pm 1.6}$	$89.61_{\pm 1.3}$	$83.98_{\pm 2.1}$
UJADA	$63.08_{\pm 2.8}$	$63.03_{\pm 2.7}$	$48.09_{\pm 2.9}$	$45.22_{\pm 2.6}$	$60.61_{\pm 2.7}$	$51.68_{\pm 3.2}$	$91.44_{\pm 0.9}$	$89.41_{\pm 1.1}$	$83.97_{\pm 1.7}$
BCAN	$49.32_{\pm 5.9}$	$72.07_{\pm 0.7}$	$52.90_{\pm 1.1}$	$38.69_{\pm 3.5}$	$61.20_{\pm 2.4}$	$51.25_{\pm 3.7}$	$89.86_{\pm 2.0}$	$89.18_{\pm 2.9}$	$83.62_{\pm 4.2}$
PASDA	$46.81_{\pm 5.2}$	$67.88_{\pm 1.1}$	$50.41_{\pm 1.7}$	$46.21_{\pm 3.4}$	$62.24_{\pm 0.9}$	$52.71_{\pm 1.1}$	$93.68_{\pm 1.7}$	$92.68_{\pm 2.4}$	$88.80_{\pm 3.5}$
SCLUDA	$59.37_{\pm 5.2}$	$74.39_{\pm 2.0}$	$56.38_{\pm 3.2}$	$33.29_{\pm 2.6}$	$60.17_{\pm 0.9}$	$49.16_{\pm 1.4}$	$91.66_{\pm 0.6}$	$90.33_{\pm 0.6}$	$85.23_{\pm 0.9}$
JAN-STF	$62.02_{\pm 2.4}$	$70.84_{\pm 3.0}$	$55.51_{\pm 2.8}$	$40.90_{\pm 4.6}$	$60.13_{\pm 1.2}$	$50.24_{\pm 1.9}$	$92.29_{\pm 0.6}$	$91.05_{\pm 0.8}$	$86.30_{\pm 1.2}$
MLUDA	$62.47_{\pm 4.1}$	$75.26_{\pm 2.8}$	$58.11_{\pm 3.5}$	$35.13_{\pm 2.1}$	$59.66_{\pm 0.9}$	$48.76_{\pm 0.6}$	$91.67_{\pm 0.7}$	$90.00_{\pm 0.9}$	$84.76_{\pm 1.3}$
CSGNet	$53.37_{\pm 5.1}$	$73.95_{\pm 2.2}$	$55.97_{\pm 3.2}$	$48.25_{\pm 3.8}$	$60.15_{\pm 0.5}$	$50.00_{\pm 1.0}$	$93.46_{\pm 0.7}$	$92.37_{\pm 0.7}$	$88.31_{\pm 1.1}$
SGLPCL	$\mathbf{63.68}_{\pm \mathbf{4.2}}$	$\mathbf{77.48}_{\pm \mathbf{1.5}}$	$\mathbf{62.43}_{\pm \mathbf{2.3}}$	$\mathbf{48.84}_{\pm \mathbf{4.0}}$	$\mathbf{64.44}_{\pm \mathbf{0.8}}$	$\mathbf{55.98}_{\pm \mathbf{1.4}}$	$\mathbf{95.26}_{\pm \mathbf{0.4}}$	$\mathbf{94.67}_{\pm \mathbf{0.6}}$	$\mathbf{91.78}_{\pm \mathbf{1.0}}$

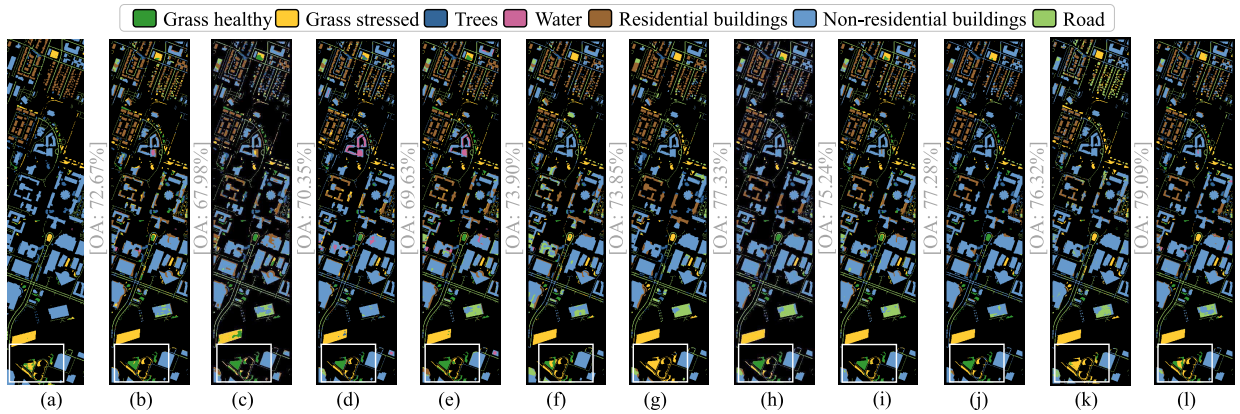


Fig. 8. Classification maps of 11 methods for the target domain (Houston2018) in the Houston dataset. (a) GT, (b) JADA, (c) BCDM, (d) TSTnet, (e) UJADA, (f) BCAN, (g) PASDA, (h) SCLUDA, (i) JAN-STF, (j) MLUDA, (k) CSGNet, and (l) SGLPCL.

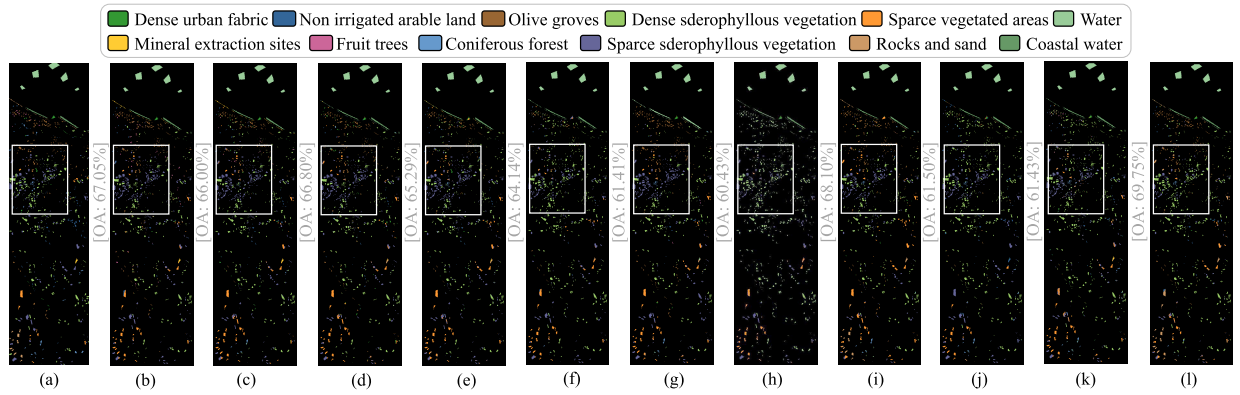


Fig. 9. Classification maps of 11 methods for the target domain (Loukia) in the HyRANK dataset. (a) GT, (b) JADA, (c) BCDM, (d) TSTnet, (e) UJADA, (f) BCAN, (g) PASDA, (h) SCLUDA, (i) JAN-STF, (j) MLUDA, (k) CSGNet, and (l) SGLPCL.

2) Qualitative Analysis: Figs. 8–10 present the ground truth (GT) and classification maps generated by SGLPCL and ten leading methods across the Houston, HyRANK, and S–H datasets, providing a visual comparison of model generalization and spatial coherence. Fig. 8 illustrates the GT and corresponding classification maps, demonstrating that SGLPCL produces spatially refined and semantically coherent predictions that align closely with actual land-cover distributions. Within the highlighted white region, while most comparison methods fail to distinguish between stressed and healthy grass due to spectral similarity, both SGLPCL and BCAN achieve accurate separation. However, BCAN exhibits confusion between nonresidential buildings and roads—a limitation effectively mitigated by SGLPCL through its pseudolabel refinement strategy guided by robust source-domain supervision. Nevertheless, nonresidential buildings display substantial variability across locations, posing challenges for pseudolabel refinement. For instance, in the upper-right portion of the white-bordered region, SGLPCL misclassifies some nonresidential buildings as roads, which in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

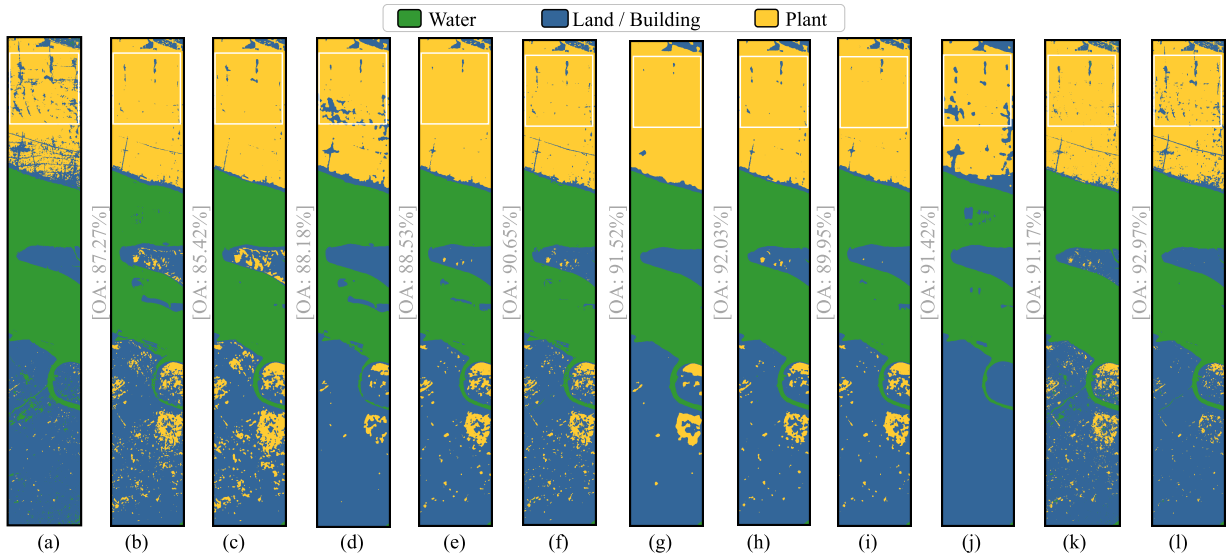


Fig. 10. Classification maps of ten methods for the target domain (Shanghai) in the S–H dataset. (a) GT. (b) JADA, (c) BCDM, (d) TSTnet, (e) UJADA, (f) BCAN, (g) PASDA, (h) SCLUDA, (i) JAN-STF, (j) MLUDA, (k) CSGNet, and (l) SGLPCL.

turn leads to a reduction in the road class's accuracy. As shown in Fig. 9, the HyRANK dataset poses a greater challenge due to its inherently complex category distribution and fine-grained vegetation types, leading to notable noise when transferring knowledge from the source domain (Loukia). In the white-bordered region, most baseline algorithms misclassify dense sclerophyllous vegetation as sparse sclerophyllous vegetation, whereas SGLPCL and SCLUDA consistently enhance category discrimination through contrastive feature alignment, yielding more reliable predictions. Moreover, in the upper portion of the marked region, SCLUDA confuses olive groves with fruit trees, while SGLPCL—benefiting from global prototype-level contrast and the additional contextual integration of nearby coastal water—achieves more accurate semantic consistency with GT. On the S–H dataset (Fig. 10), similar to most comparison methods, SGLPCL occasionally misclassifies the lower portion of the image, confusing Water and Land/Building with Plant due to vegetation diversity and spectral discrepancy. However, within the highlighted white region, only SGLPCL successfully distinguishes Land/Building from Plant, producing classification results that most closely align with the GT. Collectively, these observations confirm the model's superior capability in preserving spatial boundaries and semantic integrity across diverse and complex domain shifts.

E. Ablation Study

1) Overall Impact of the Three Components: To thoroughly assess the effectiveness of each proposed component, we perform ablation experiments to analyze the individual and combined contributions of the three core modules: GAT-3SE (G), ADFS (A), and SATS (S). These experiments allow for a detailed evaluation of the impact of each module both independently and in conjunction with the others. The corresponding results are presented in Fig. 11.

In Fig. 11, "w/o G, A, and S" denotes excluding all components, while "w/ G only" indicates using only

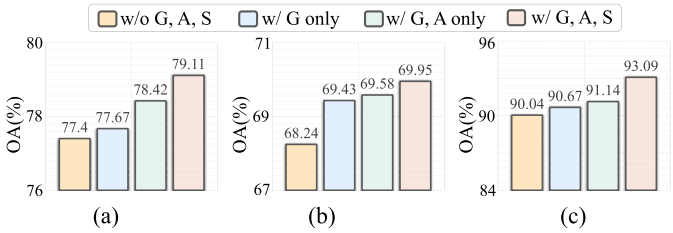


Fig. 11. Ablations of three components on OA (%). (a) Houston. (b) HyRANK. (c) S–H.

GAT-3SE, and so on. Fig. 11 shows that adding GAT-3SE yields consistent performance gains across all datasets, with OA improvements of 1.57%, 2.55%, and 0.63% on Houston, HyRANK, and S–H, respectively. This verifies the importance of global representation learning for domain-level feature alignment. By leveraging spectral–spatial graph reasoning and self-enhancement, GAT-3SE captures transferable structural dependencies, reducing discrepancies between heterogeneous domains. Building on this, introducing ADFS further improves performance by 0.75%, 0.15%, and 0.47%. This gain arises from adaptive pseudolabel refinement that incorporates source supervision into label generation, reducing noise and stabilizing optimization, especially under large domain shifts. Finally, integrating SATS yields additional OA gains of 0.69%, 0.37%, and 1.95%, highlighting its role in stable prototype construction and cross-domain semantic calibration. Through prototype-level contrastive alignment with temporal smoothing, SATS produces a more robust feature space that preserves class-level semantics during adaptation. These results demonstrate the complementary synergy among GAT-3SE, ADFS, and SATS, whose joint optimization unifies global representation learning, reliable pseudolabel refinement, and semantic-consistent prototype alignment, leading to robust performance gains.

2) Detailed Analysis of Core Designs: Building upon the component-level ablation study, we further conduct

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

ABLATIONS OF THE CORE DESIGNS IN THE PROPOSED SGLPCL ACROSS THREE DATASETS (HOUSTON, HYRANK, AND S–H) WITH OA (%)

EXP	Ablation	Houston	HyRANK	S-H
1	w/o 3SE	78.45	68.25	91.10
2	w/o H3MD	77.73	69.41	90.01
3	w/o Refinement	78.23	68.91	90.95
4	w/o ATS	78.38	69.05	91.82
5	w/o ATS+Refinement	77.70	68.62	90.58
6	w/o Sc	78.63	68.42	91.68
7	w/ Global-PL only	77.67	69.50	90.74
8	w/ Local-PL only	78.87	69.67	92.04
9	w/ Euclidean only	79.00	69.30	91.06
10	w/ MMD only	76.91	66.00	88.16
	Ours	**79.11**	**69.95**	**93.09**

fine-grained ablations to assess the impact of key algorithmic designs within SGLPCL. These studies aim to reveal how each individual design choice contributes to overall adaptation performance and stability. Ten variants are evaluated to examine key aspects such as pseudolabel refinement and prototype construction. The configurations and results are summarized in Table VII, where w/o and w/ indicate "without" and "with."

Table VII demonstrates that excluding 3SE node construction and using only spectral features (EXP-1) leads to OA drops of 0.66%, 1.70%, and 1.99% on Houston, HyRANK, and S–H, respectively. This consistent degradation across all datasets underscores the indispensable role of the 3SE module in capturing complementary spatial and semantic dependencies beyond spectral cues, which is particularly crucial for complex urban and heterogeneous regions. When removing the holistic memory MMD (H3MD) criterion (EXP-2), OA decreases by 1.38%, 0.54%, and 3.08% across the three datasets, indicating that H3MD effectively mitigates interdomain feature shifts by enforcing global distributional alignment. The more significant performance drop on S–H further confirms that H3MD is especially beneficial for domains with larger spectral discrepancies.

Eliminating the memory pool refinement process (EXP-3) results in OA reductions of 0.88%, 1.04%, and 2.14%, demonstrating that iterative prototype refinement improves the reliability of pseudolabels and stabilizes the cross-domain representation learning process. Without the ATS calibration (EXP-4), OA decreases by 0.73%, 0.90%, and 1.27%, suggesting that the adaptive similarity adjustment enhances feature discrimination and reduces false alignment among semantically close classes. Comparing the removal of both ATS and Refinement (EXP-5), the results of EXP-3 still outperform EXP-5 by 0.53%, 0.29%, and 0.37%, demonstrating that ATS effectively calibrates class prototypes even when pseudolabel quality is degraded due to the absence of refinement. When the source centroid (SC) is excluded from target prototype generation (EXP-6), OA drops by 0.48%, 1.53%, and 1.41%, which validates the necessity of source-guided prototype correction in (10) for maintaining cross-domain consistency.

Moreover, using only global or local pseudolabels (EXP-7, 8) causes OA decreases of 1.44%, 0.45%, 2.35% and 0.24%, 0.28%, 1.05%, respectively. This complementary behavior confirms that global pseudolabels offer high-level semantic stability, while local pseudolabels contribute to finer spatial adaptation; their joint optimization provides balanced supervision that enhances both label diversity and local coherence. Finally, employing only Euclidean or MMD distance (EXP-9, 10) for pseudolabel refinement reduces OA by 0.11%, 0.65%, 2.03% and 2.20%, 3.95%, 4.93%, respectively. This difference indicates that Euclidean distance captures precise instance-level similarity, whereas MMD focuses on distribution-level consistency; combining them yields a complementary effect that jointly strengthens instance precision and global robustness across domains. Overall, no single design alone drives the entire algorithm's performance; each core component plays a crucial role in the overall effectiveness of SGLPCL.

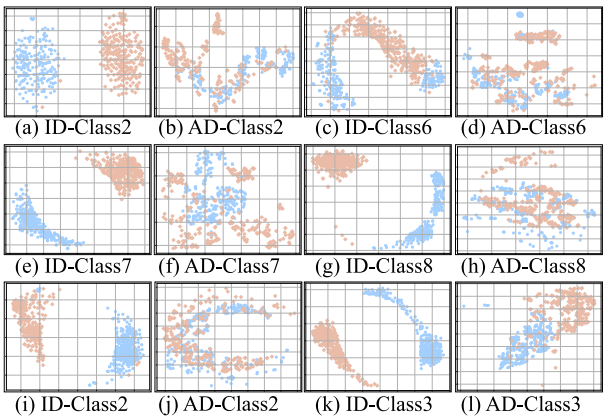


Fig. 12. Alignment performance of SGLPCL across three datasets. (a)–(d) Houston, (e)–(h) HyRANK, and (i)–(l) S–H. Blue circles and red diamonds represent samples from the source and target domains, respectively.

F. Alignment Performance

To evaluate the cross-domain alignment capability of SGLPCL, we adopt t-SNE [51] to visualize backbone-extracted high-dimensional features. Fig. 12 shows both the initial distributions (IDs) and the aligned distributions (ADs) after applying SGLPCL, providing an intuitive illustration of its effectiveness in mitigating domain shift.

Fig. 12(a), (c), (e), (g), (i), and (k) depict the initial cross-domain feature distributions on the Houston, HyRANK, and S–H datasets. Even within the same semantic class, features from the source and target domains are scattered and often separated, indicating significant domain discrepancy. After applying SGLPCL with global–local prototype contrastive learning under source-domain supervision, the intraclass consistency is significantly improved, as reflected in the ADs shown in Fig. 12(b), (d), (f), (h), (j), and (l). Features belonging to the same class from different domains are pulled closer, while interclass separability is maintained, demonstrating the effectiveness of the proposed alignment strategy. For example, in the Houston dataset, Fig. 12(a) and (b) illustrate that class 2 features, which were initially dispersed across source and target domains, achieve near-complete overlap after SGLPCL. Similar trends are observed in the HyRANK and S–H datasets, where the ADs show clearer clustering and reduced cross-domain divergence. Overall, these results provide strong qualitative evidence that SGLPCL effectively

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
TIME RESULTS FOR NINE METHODS ON THE HOUSTON DATASET

Metric	T	U	B	P	S	J	M	C	Ours
TT (s)	**1.23**	1.97	1.75	2.29	1.89	1.25	2.05	10.7	6.95
IT (s)	**0.12**	0.13	0.13	1.41	1.68	0.12	4.18	4.42	1.23
OA (%)	70.3	69.4	73.2	73.6	77.0	75.2	77.0	75.9	**79.1**

bridges the domain gap, producing a feature space that is both semantically discriminative and domain-consistent.

G. Time Complexity Analysis

SGLPCL consists of three main stages: GAT-3SE, ADFS, and SATS. The computational cost of GAT-3SE is dominated by multihead graph attention operations performed independently on source and target node sets. Let n_s and n_t denote the numbers of nodes in the source and target graphs, respectively. Each node embedding of dimension d_h is projected into a d'-dimensional latent space and attends to a sparsely sampled neighborhood of size d ($d \ll n_s, n_t$) following [16]. For each attention head, the per-node complexity is $\mathcal{O}(d \cdot d')$, and with K attention heads, the overall complexity of GAT-3SE is $\mathcal{O}(K \cdot d \cdot d' \cdot (n_s + n_t))$, which scales linearly with the number of nodes. The computational overhead of ADFS arises from the memory pool update mechanism and MMD-based distance calculations, which are applied to global and local features in an identical manner. Let n denote the memory pool capacity, n_{top} the number of nearest neighbors selected from each memory pool, and d' the dimensionality of global representations. For a single target feature g_m^t, memory pool updates incur an $\mathcal{O}(d')$ cost, while neighbor retrieval requires $\mathcal{O}(n \cdot d')$ operations. The intradomain MMD terms computed over the top-n_{top} neighbors have complexity $\mathcal{O}(n_{top}^2 \cdot d')$, and the holistic cross-domain MMD term introduces an additional $\mathcal{O}(n^2 \cdot d')$ cost. Since n and n_{top} are fixed hyperparameters independent of dataset size, the per-sample complexity simplifies to $\mathcal{O}(d')$, leading to an overall complexity of $\mathcal{O}(n_t \cdot d')$ for refining all target samples. The same order of complexity applies to the local-feature refinement process. Finally, SATS computes pairwise cosine similarities between source and target prototypes, with complexity $\mathcal{O}(c^2)$, where c is the number of classes. Overall, the computational complexity of SGLPCL scales linearly with the number of samples, with the GAT-3SE module being the dominant cost. ADFS and SATS introduce only moderate overhead in comparison.

To provide a clear comparison of time complexity, we evaluate SGLPCL against eight HSDA methods—TSTnet (T), UJADA (U), BCAN (B), PASDA (P), SCLUDA (S), JAN-STF (J), MLUDA (M), and CSGNet (C)—on the Houston dataset under consistent settings. Training time (TT), inference time (IT), and OA are reported, where TT and IT are averaged per epoch. Results are shown in Table VIII.

According to the results in Table VIII, SGLPCL achieves the highest OA (79.1%) among the nine methods, outperforming CSGNet by 3.2%. It also demonstrates strong time efficiency, with a TT of 6.95 s and IT of 1.23 s, both significantly lower than CSGNet's 10.7 and 4.42 s. Other methods, such as TSTnet and JAN-STF, exhibit shorter training and ITs but at the cost of lower accuracy (70.3% and 75.2%, respectively). These results highlight SGLPCL's ability to effectively balance computational efficiency and accuracy in HSDA.

IV. CONCLUSION

HSDA has traditionally viewed source supervision as a secondary stabilizer, often overlooking global alignment, which leads to unstable transfer in label-scarce and spectrally–spatially complex scenarios. To address this, we propose SGLPCL, a source-driven framework that centers source supervision as the core adaptation driver. SGLPCL refines domain alignment through three stages: GAT-3SE encodes spectral–spatial–semantic cues for global representation learning, ADFS injects source guidance for pseudolabel refinement and prototype formation, and SATS constructs source-guided prototypes with ATS, improving intraclass compactness and interclass separability. Experiments on the Houston, HyRANK, and S–H datasets demonstrate the clear superiority of SGLPCL over ten state-of-the-art methods. Under label-scarce settings (5% source labels), SGLPCL shows strong robustness across both VGG16 and deep residual 3DCNN backbones, achieving OA gains of (2.06%, 2.22%), (1.96%, 1.08%), and (1.09%, 2.30%) on the three datasets, respectively. Moreover, across varying source sampling ratios, SGLPCL maintains consistent improvements—rising from 81.5% (10%) to 83.0% (100%) on Houston and outperforming competitors by 2.38%–3.16% and 0.34%–1.75% on HyRANK and S–H, respectively. Ablation studies validate the effectiveness of all three modules, and feature distribution visualizations confirm SGLPCL's strong cross-domain alignment.

While SGLPCL generalizes well in challenging HSDA scenarios, several aspects merit further study. Although ATS enables effective interdomain alignment, more dynamic or instance-aware variants may improve adaptability. Moreover, advanced pseudolabel refinement could better suppress residual noise in low-density target domains.

REFERENCES

[1] F. Wang, J. Li, Q. Yuan, and L. Zhang, "Local–global feature-aware transformer based residual network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546119.

[2] R. Shang et al., "Hyperspectral image classification based on pyramid coordinate attention and weighted self-distillation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544316.

[3] X. Wang, K. Tan, Q. Du, Y. Chen, and P. Du, "CVA2E: A conditional variational autoencoder with an adversarial training process for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5676–5692, Aug. 2020.

[4] W. Zhang, M. Hu, S. Hou, R. Shang, J. Feng, and S. Xu, "Group-spectral superposition and position self-attention transformer for hyperspectral image classification," *Expert Syst. Appl.*, vol. 265, Mar. 2025, Art. no. 125846.

[5] L. Zhang, L. Zhang, D. Tao, X. Huang, and B. Du, "Hyperspectral remote sensing image subpixel target detection based on supervised metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4955–4965, Aug. 2014.

[6] W. Li, W. Wei, L. Zhang, C. Wang, and Y. Zhang, "Unsupervised deep domain adaptation for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Yokohama, Japan, Jul. 2019, pp. 1–4.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.

[7] R. Xu, P. Liu, L. Wang, C. Chen, and J. Wang, "Reliable weighted optimal transport for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4394–4403.

[8] J. Zhu, H. Bai, and L. Wang, "Patch-mix transformer for unsupervised domain adaptation: A game perspective," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3561–3571.

[9] M. Long, Z. Cao, J. Wang, and M. I. Jordan, "Conditional adversarial domain adaptation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–11.

[10] S. Li, F. Lv, B. Xie, C. H. Liu, J. Liang, and C. Qin, "Bi-classifier determinacy maximization for unsupervised domain adaptation," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 10, pp. 8455–8464.

[11] S. Li, C. H. Liu, B. Xie, L. Su, Z. Ding, and G. Huang, "Joint adversarial domain adaptation," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 729–737.

[12] W. Li et al., "Few-shot learning based on embedded self-distillation and adaptive Wasserstein distance for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5502315.

[13] H. Lee, S. Eum, and H. Kwon, "Cross-domain CNN for hyperspectral image classification," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 3627–3630.

[14] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9646–9660, Nov. 2021.

[15] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, "A kernel two-sample test," *J. Mach. Learn. Res.*, vol. 13, no. 1, pp. 723–773, 2012.

[16] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.

[17] R. Shang, Y. Xie, W. Zhang, J. Feng, and S. Xu, "Joint adversarial network with semantic and topology fusion for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5521116.

[18] B. Cui, S. Huang, G. Ren, H. Li, and J. Yu, "Category semantic-guided unsupervised domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5512016.

[19] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 59, pp. 1–35, 2016.

[20] X. Tang, C. Li, and Y. Peng, "Unsupervised joint adversarial domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536415.

[21] Z. Li et al., "Cycle self-training with joint adversarial for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5531117.

[22] K. Mei, C. Zhu, J. Zou, and S. Zhang, "Instance adaptive self-training for unsupervised domain adaptation," in *Proc. 16th Eur. Conf. Comput. Vis.-ECCV*, 2020, pp. 415–430.

[23] S. Zhong and Y. Zhang, "An iterative training sample updating approach for domain adaptation in hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 10, pp. 1821–1825, Oct. 2021.

[24] Z. Feng, S. Tong, S. Yang, X. Zhang, and L. Jiao, "Pseudo-label-assisted subdomain adaptation for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 6, pp. 4729–4744, Jun. 2024.

[25] H. Wang, Y. Cheng, X. Liu, and Y. Kong, "Bi-classifier adversarial network for cross-scene hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[26] Y. Li, L. Guo, and Y. Ge, "Pseudo labels for unsupervised domain adaptation: A review," *Electronics*, vol. 12, no. 15, p. 3325, Aug. 2023.

[27] Q. Liu, J. Peng, G. Zhang, W. Sun, and Q. Du, "Deep contrastive learning network for small-sample hyperspectral image classification," *J. Remote Sens.*, vol. 3, p. 25, Jan. 2023.

[28] Z. Li et al., "Supervised contrastive learning-based unsupervised domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524017.

[29] Y. Ning, J. Peng, Q. Liu, Y. Huang, W. Sun, and Q. Du, "Contrastive learning based on category matching for domain adaptation in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5301814.

[30] M. Cai et al., "Mind the gap: Multilevel unsupervised domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5520014.

[31] Y. Wang, X. Yu, H. Dong, and S. Zang, "Graph convolutional network with local and global feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5536915.

[32] H. Tang, Y. Wu, H. Li, D. Tang, X. Yang, and W. Xie, "Global–local prototype-based few-shot learning for cross-domain hyperspectral image classification," *Knowl.-Based Syst.*, vol. 314, Apr. 2025, Art. no. 113199.

[33] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–7.

[34] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.*, vol. 27, no. 3, pp. 379–423, Jul. 1948.

[35] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.

[36] X. Zhu, Z. Ghahramani, and J. D. Lafferty, "Semi-supervised learning using Gaussian fields and harmonic functions," in *Proc. 20th Int. Conf. Mach. Learn. (ICML)*, 2003, pp. 912–919.

[37] M. Litrico, A. Del Bue, and P. Morerio, "Guiding pseudo-labels with uncertainty estimation for source-free unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7640–7650.

[38] I. Misra and L. Van Der Maaten, "Self-supervised learning of pretext-invariant representations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6707–6717.

[39] Y. Kim, J. Yim, J. Yun, and J. Kim, "NLNL: Negative learning for noisy labels," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 101–110.

[40] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[41] B. Le Saux, N. Yokoya, R. Hansch, and S. Prasad, "2018 IEEE GRSS data fusion contest: Multimodal land use classification [technical committees]," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 1, pp. 52–54, Mar. 2018.

[42] K. Karantzalos, C. Karakizi, Z. Kandylakis, and G. Antoniou, "HyRANK hyperspectral satellite dataset I (version v001)," Zenodo, Remote Sens. Lab., Nat. Tech. Univ. Athens, Athens, Greece, Tech. Rep., Apr. 2018, doi: 10.5281/zenodo.1222202.

[43] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–12.

[44] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[45] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[46] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Spectral–spatial hyperspectral image segmentation using subspace multinomial logistic regression and Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 809–823, Mar. 2012.

[47] Z. Wu, Y. Xiong, S. X. Yu, and D. Lin, "Unsupervised feature learning via non-parametric instance discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3733–3742.

[48] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[49] W. S. Gosset"The probable error of a mean", *Biometrika*, pp. 1–25, 1908.

[50] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[51] L. Van Der Maaten, "Accelerating t-SNE using tree-based algorithms," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 3221–3245, 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:04:06 UTC from IEEE Xplore. Restrictions apply.